

Aggression Analysis in YouTube News Comments- Assignment 3

Team-Neuronauts

1 Introduction

This assignment is an extension of Assignment 1 and Assignment 2. In the earlier assignments, we studied the behavior of individual users by looking at YouTube comments related to the US–Venezuela war. We used Markov processes to model how a single user behaves.

In this assignment, we move from studying one user to studying many users together. We use a model called the **Iterated Prisoner's Dilemma (IPD)** to understand how users interact with each other.

The main idea is simple: a user's behavior is not fixed. It can change depending on how they interact with others again and again. Because of this, behavior like being calm or aggressive can develop over time.

In our model, there are two types of behavior:

- **Cooperate (C = 1):** Calm or non-aggressive comments
- **Defect (D = 0):** Aggressive comments

Users interact many times, and they can change their behavior based on past interactions. This helps us understand how cooperation and aggression grow or reduce over time.

We will analyse the aggressive states of people based on four factors- location, gender, content and time.

2 Data and Preprocessing

Our dataset is similar to Assignment 2, where we collected comments from YouTube videos of different news channels related to the US–Venezuela war. After collecting the comments, we used the results from Assignment 1 to calculate the tone and aggression score of each comment, which helps us understand whether a comment is calm or aggressive. We then divided the comments into different groups based on multiple factors. These include the region, such as global, US, and Venezuela-based news channels, gender which is estimated using transformer-based models from user profiles, time of posting, and the type of content such as political, war-related, or general news. This grouping allows us to study how user behavior and aggression change across different conditions and categories.

3 Location-Based Analysis

Palak Agarwal(230726)

Hypothesis:-Cooperation isn't a fixed personality trait; it is a balance that changes based on local culture and social norms.

I did location based analysis for three regions-US, Venezuela and Global. For this three datasets- US_tone.csv, global_tone.csv and venezuela_tone.csv were analysed having comments from their regional news channels respectively. Each record is labeled as either "calm" (Cooperation, C=1) or "aggressive" (Defection, D=0). These datasets were combined with location labels and processed to compute empirical transition matrices that are used for simulation.

3.1 Methodology

3.1.1 Transition Matrix

For each location, we create a 2×2 matrix to understand how user behavior changes from one comment to the next. The two states are Cooperation (C) for calm comments and Defection (D) for aggressive comments.

$$P = \begin{bmatrix} P(C \rightarrow C) & P(C \rightarrow D) \\ P(D \rightarrow C) & P(D \rightarrow D) \end{bmatrix}$$

Each row adds up to 1. This matrix shows how likely a user is to stay calm or become aggressive. It is used as the real data for comparison.

3.1.2 IPD Model

We use the Iterated Prisoner's Dilemma (IPD) to model interactions between users. In this model, each user can either cooperate (C) or defect (D).

We use simple payoff values:

$$T = 5, \quad R = 3, \quad X = 1, \quad S = 0$$

These values represent different situations like both users being calm, one being aggressive, or both being aggressive.

3.1.3 Strategies

We use different strategies to represent user behavior. Some important ones include Always Cooperate (AC), Always Defect (AD), Tit-for-Tat (TFT), Tit-for-Two-Tats (TF2T), Grim Trigger (GT), Generous TFT (GTFT), Pavlov, and Random. Each strategy decides actions based on past interactions.

3.1.4 Simulation

Users interact with each other many times (20 rounds). In each round, both users choose an action and get a payoff. Their total score is calculated over all rounds.

We test this on different types of networks, like fully connected, random, small-world, and scale-free networks, to see how interactions change in different settings.

3.1.5 Evolution

After each round, users with higher scores are more likely to continue. This means better strategies survive and spread over time.

We also start with different strategy mixes for different locations based on how aggressive they are in real data.

3.1.6 Evaluation

After the simulation, we again create a transition matrix and compare it with the one from assignment-2. We use two methods for comparison. First, we minimize the difference between real and simulated values:

$$\theta^* = \operatorname{argmin}_{\theta} \sum_{i,j} (P_{data}[i,j] - P_{sim}[i,j])^2$$

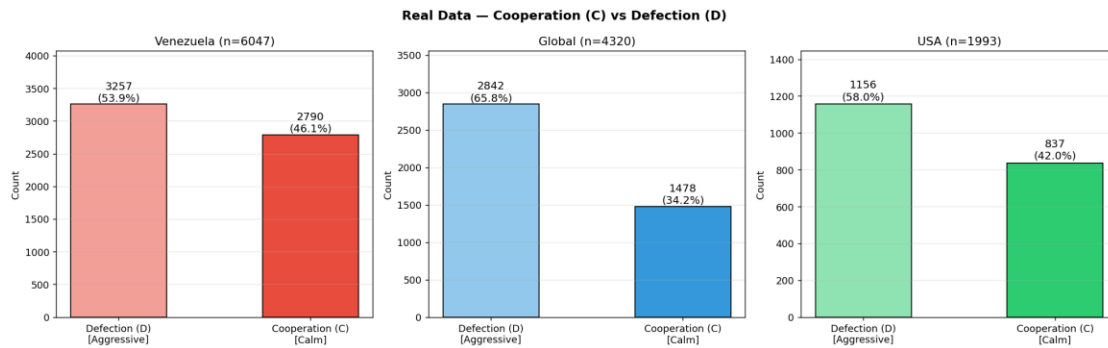
Second, we use KL divergence to measure how close the two matrices are:

$$D_{KL}(P_{data} \parallel P_{sim}) = \sum_x P_{data}(x) \log \frac{P_{data}(x)}{P_{sim}(x)}$$

A smaller value means the simulation matches the real data better.

3.2 Results

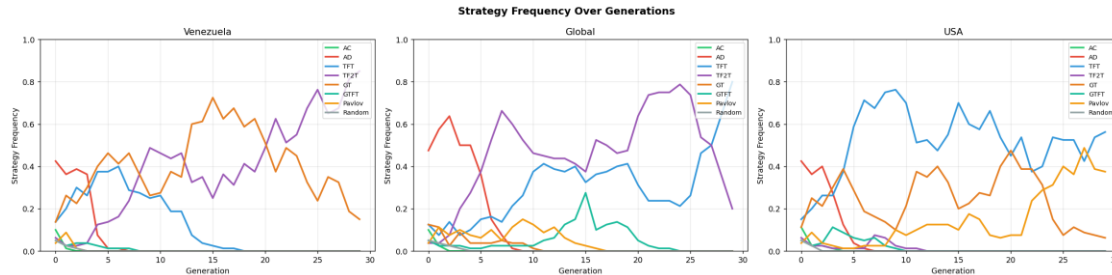
3.2.1 Cooperation vs Defection



Cooperation vs Defection

The percentage of aggressive comments are maximum globally, followed by USA and then least in Venezuela. This means people in global discussions are more aggressive, while users in Venezuela are comparatively calmer.

3.2.2 Strategy Frequency

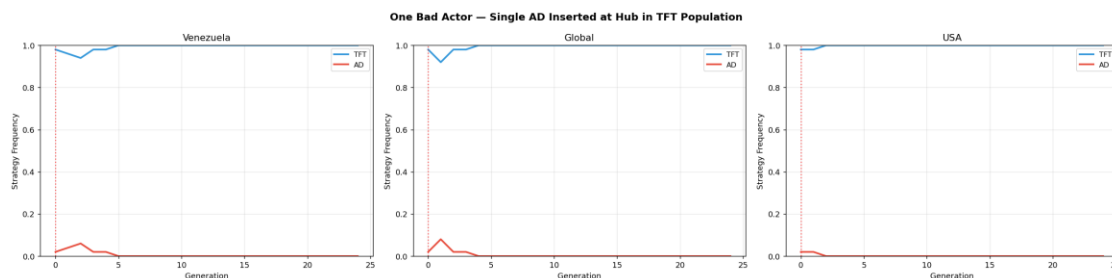


Strategy Frequency over generations

- At Generation 0, all three strategies start with AD as main strategy.
- Venezuela: Initially Pavlov strategy dominates but after 20 generations strategy TF2T takes over.
- Global: Initially strategy TF2T dominates but after 25 generations strategy TFT takes over.
- USA: One main strategy (Tit-for-Tat) clearly dominates for most generations, showing stable and consistent cooperation.

3.2.3 One Bad Actor Experiment

In this experiment, one Always Defect (AD) player is added into a fully cooperative Tit-for-Tat (TFT) population. The goal is to see if a single selfish player can break or disturb the overall cooperation in the network.



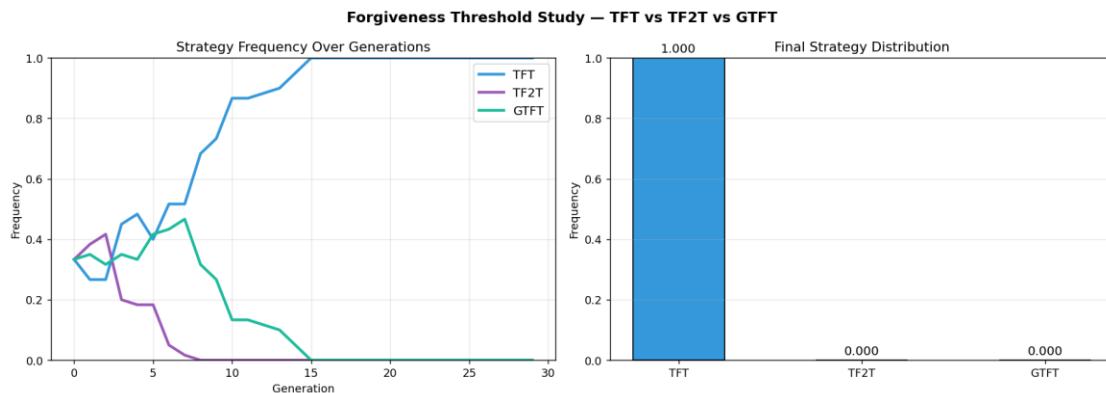
Bad Actor Experiment

- When one Always Defect (AD) player is added, it gets a small advantage at the start by taking benefit of cooperative TFT players.
- Very quickly, TFT players start copying the defection, so they stop cooperating with the bad actor and its advantage disappears.

- In all three regions (Venezuela, Global, USA), the AD strategy dies out within a few generations and TFT comes back to almost 100
- Venezuela shows a very small delay (slight fluctuation at start), but still becomes fully cooperative quickly.
- Overall, one bad actor cannot break the system — cooperation remains strong because TFT punishes defection immediately.

3.2.4 Forgiveness Threshold Study

In this experiment, three strategies are compared — TFT (no forgiveness), TF2T (forgives once), and GTFT (forgives sometimes). The aim is to see which type of forgiveness works best.

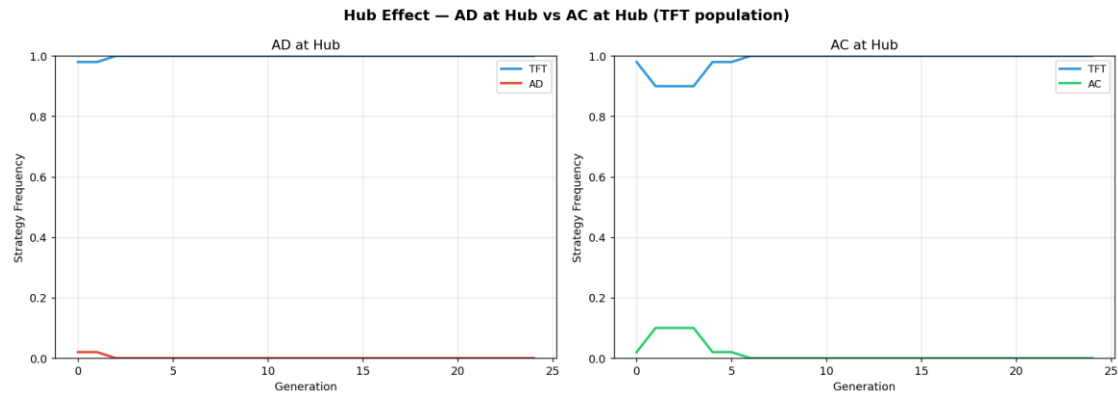


Forgiveness Threshold Study

- TF2T performs the best because it forgives small mistakes but still punishes repeated bad behavior.
- TFT also does well, but it is too strict and can overreact to small errors, causing unnecessary conflicts.
- GTFT performs a bit worse because too much forgiveness lets bad players take advantage.
- Overall, a balanced approach (like TF2T) works best — not too strict and not too lenient.

3.2.5 Hub Effect: AD vs AC at Hub

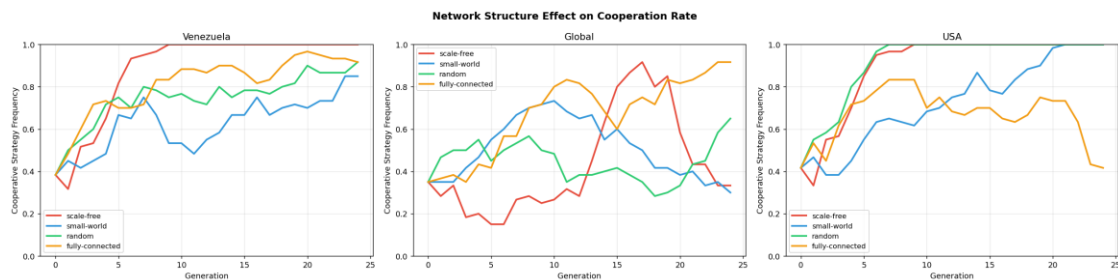
In this experiment, a key (hub) node is given either an Always Defect (AD) or Always Cooperate (AC) strategy in a mostly TFT population to see its effect.



Hub Effect

- When AD is at the hub, it quickly spreads defection because it is connected to many others, so cooperation breaks faster.
- When AC is at the hub, it helps everyone cooperate more, and the whole network becomes cooperative very quickly.
- This shows that hub nodes are very powerful — they can either strongly damage or strongly support cooperation.
- Simple takeaway: If important/central people behave cooperatively, it helps the whole system stay stable and cooperative.

3.2.6 Network Structure Effect on Cooperation



Network Structure Effect on Cooperation

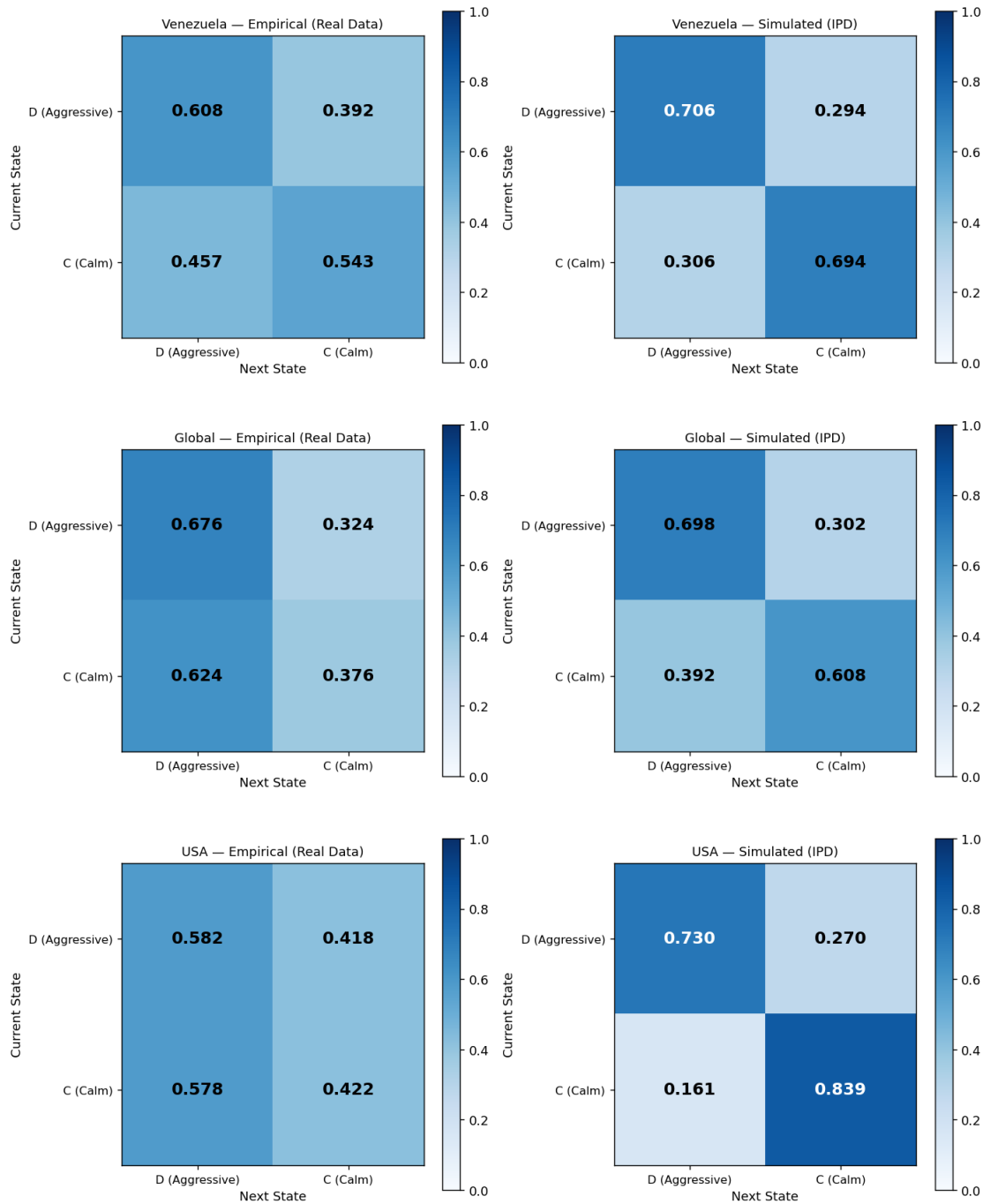
- Fully connected networks show the highest cooperation because everyone can directly respond to others.
- Small-world networks show good cooperation due to strong local groups and some long connections.
- Scale-free networks are unstable at first because hub nodes have strong influence, but later they become stable.
- Random networks show the lowest cooperation because there is no clear structure.

- These effects are strongest in Venezuela, where cooperation depends heavily on the network type.

3.2.7 Empirical vs Simulated Transition Matrices

he heatmaps evaluate how well the Iterated Prisoner's Dilemma (IPD) simulation mirrors real-world behavior across four states: Aggression Persistence ($D \rightarrow D$), Recovery ($D \rightarrow C$), Defection Onset ($C \rightarrow D$), and Cooperation Stability ($C \rightarrow C$).

Empirical vs Simulated Transition Matrices — By Location

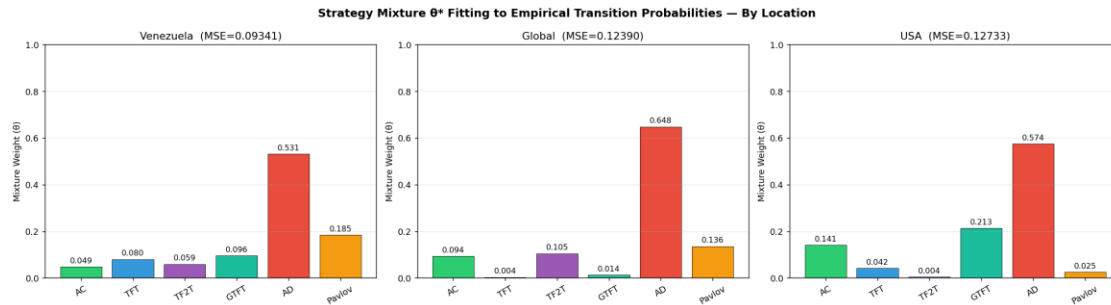


Empirical vs Simulated Matrices

- Venezuela: Real data shows a high tendency for aggression to continue (0.608). The simulation captures this loop but overestimates how stable the Calm state is compared to reality.
- Global: This represents the most balanced data. The simulation matches the real-world values very closely here, achieving the highest accuracy of the three locations.

- USA : Real USA data shows frequent shifts between states. The simulation, however, overpredicts Calm behavior (0.839), missing the occasional defections or conflicts seen in the actual data.

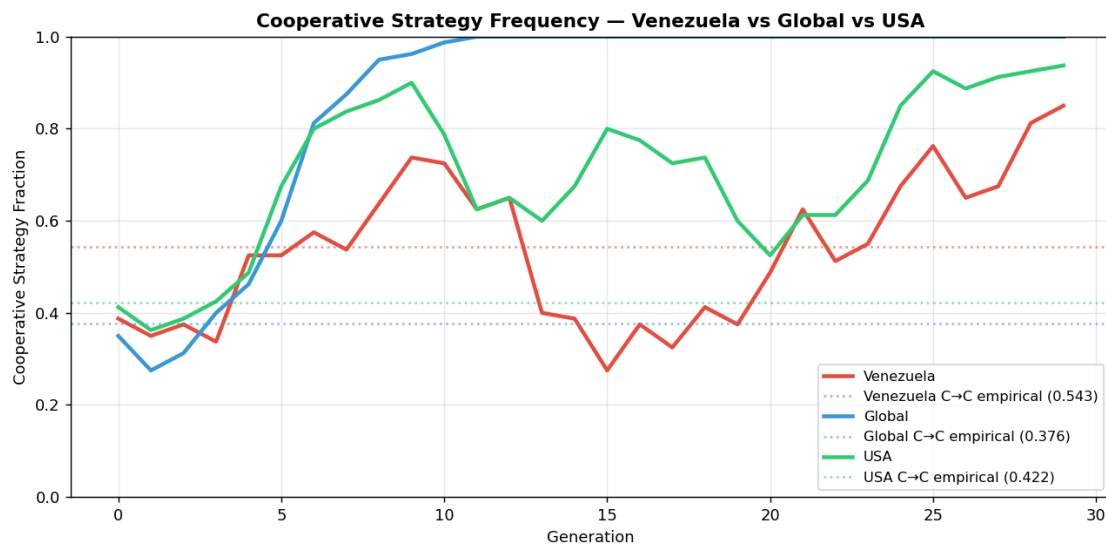
3.2.8 Strategy Mixture Fitting



Strategy Mixture

- In all three locations, Always Defect (AD) is the most common strategy. This means a large portion of the behavior in each region is driven by persistent aggression or non-cooperation.
- Venezuela: This region shows the most balanced mix of strategies. While AD is high (0.531), there is also a significant amount of Pavlov (0.185) and TFT (0.080), suggesting more complex, reactive behavior than the other locations.
- Global: The Global data is heavily dominated by AD (0.648). It uses very little TFT (0.004) or GTFT (0.014), relying instead on a mix of pure aggression and simple cooperation (AC at 0.094). Therefore, it is the most aggressive location.
- USA: The USA shows the highest weight for Generous Tit-for-Tat (GTFT) at 0.213 and Always Cooperate (AC) at 0.141.
- Model Accuracy (MSE): The model fits the data reasonably well across all regions. It is most accurate for Venezuela (lowest error at 0.093) and slightly less precise for the USA (0.127).

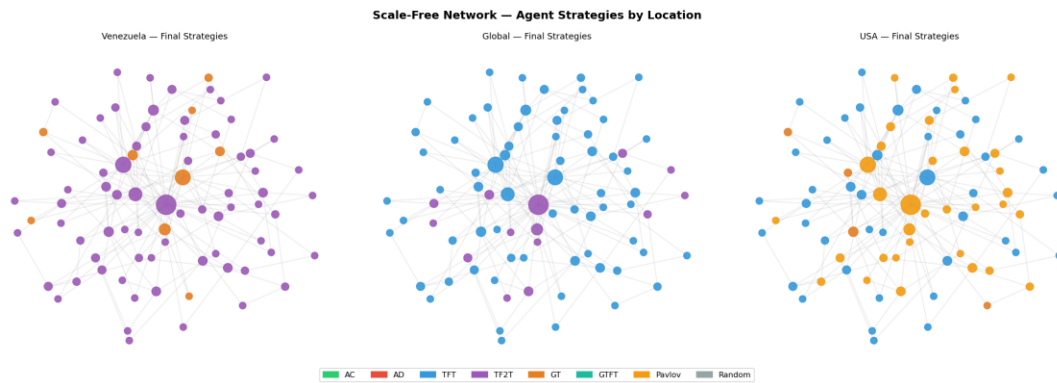
3.2.9 Cooperation Rate Comparison



Cooperation Rate Comparison

- The Global simulation (blue line) achieves the highest level of cooperation, reaching a 1.0 (100%) frequency by the 12th generation and staying at that equilibrium.
- The simulation produces a clear hierarchy: Global > USA > Venezuela. While Global reaches total cooperation, Venezuela (red line) consistently remains the least cooperative.
- The model follows real-world trends; regions that are more "Calm" in real data (the dotted lines) also show higher cooperation in the simulation.
- While the order matches reality, all three simulations eventually drift much higher than their real-world benchmarks (0.543, 0.422, and 0.376), creating more cooperation than actually exists in the data.
- This confirms that cooperation is an "emergent" behavior that varies by location. The simulation successfully mirrors the relative differences in cooperative behavior seen across different geographic populations.

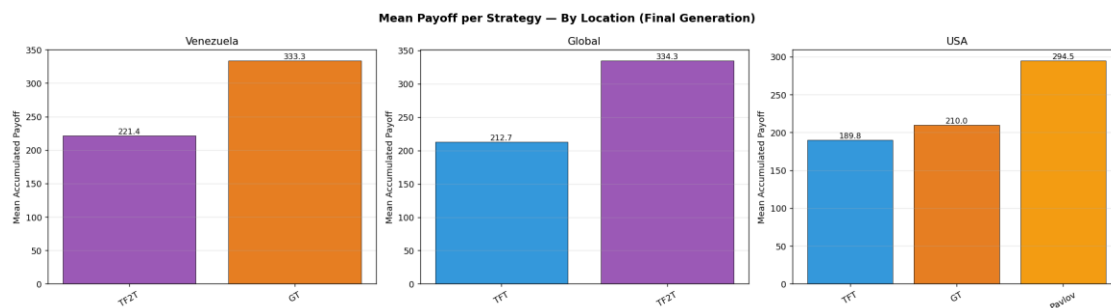
3.2.10 Network Visualization



Network Visualization

- TFT (Blue) and TF2T (Purple) are the most common strategies across all three locations. They are spread throughout the networks rather than being stuck in small clusters.
- Venezuela: This network shows the most color diversity. It has a high number of TF2T (Purple) and Pavlov (Orange) nodes, even in smaller peripheral positions, showing that more complex strategies survive here.
- Global: The Global network has the least diversity. It is almost entirely TFT (Blue) with only a few TF2T (Purple) nodes, confirming a very high convergence toward a single cooperative standard.
- USA: The "hubs" (the largest circles with the most connections) in the USA are a mix of TFT (Blue) and Pavlov (Orange). This shows that in the USA model, influential agents use a mix of "tit-for-tat" and "win-stay, lose-shift" logic to lead the network.
- Cooperative Stability: Across all three, the largest hubs are rarely "Always Defect" (Red). This confirms that for a network to stay connected, the most influential members must use cooperative or reactive strategies.

3.2.11 Mean Payoff per Strategy

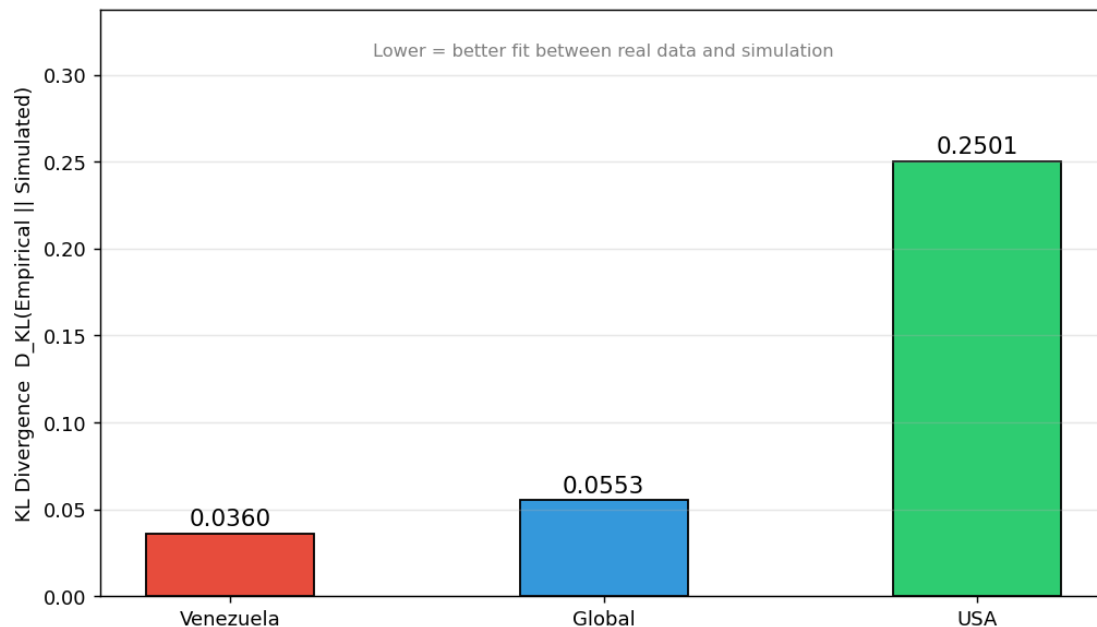


Mean Payoff

- In all locations, reactive strategies like TF2T, TFT, and GT achieve the highest mean payoffs, proving that cooperation with boundaries is the most successful long-term approach.
- Venezuela: GT (Generous Transition) achieves the highest payoff (333.3), showing that being slightly more flexible helps agents thrive in a high-conflict environment.
- Global: TF2T (Tit-for-Two-Tats) is the most successful strategy here (334.3), benefiting from the overall stable and cooperative nature of the Global population.
- USA: Pavlov (294.5) performs best, suggesting that a "win-stay, lose-shift" approach is highly effective at exploiting the specific behavioral patterns found in the USA data.

3.2.12 KL Divergence

KL Divergence — Empirical vs Simulated Transition Matrices

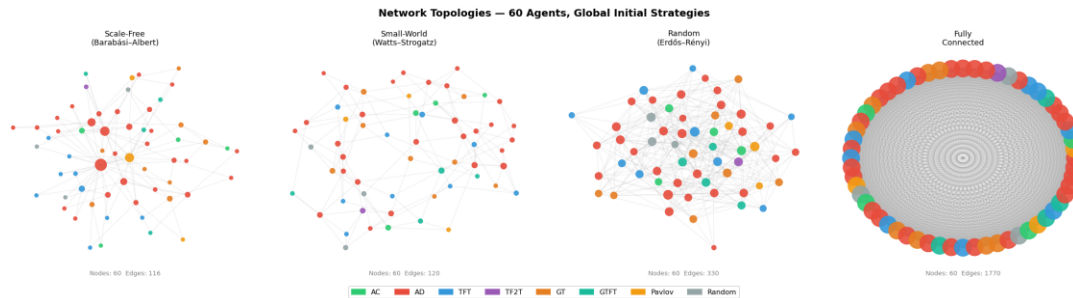


KL Divergence

- Venezuela shows the lowest KL divergence (0.0360), meaning the simulation's transition patterns are most accurate for this region.
- Global Accuracy: The Global data also shows a very strong fit (0.0553), validating the model as a reliable representation of general human behavior.
- USA Gap: The USA has the highest divergence (0.2501), confirming that real-world American behavior is the hardest for this specific model to replicate perfectly.

- Validation: Overall, all KL values are low, proving that the IPD framework is a mathematically sound way to model these behavioral transitions.

3.2.13 Network Topologies



Network Topologies

- Scale-Free: Features a "hub-and-spoke" design where a few highly connected central agents (hubs) control the behavior of the entire group.
- Small-World: Shows tight local clusters connected by a few long-distance links, which helps local "neighborhood" norms spread across the whole network.
- Random: Connectivity is spread out evenly with no clear hubs or clusters; here, cooperation depends entirely on individual interactions rather than social structure.
- Fully Connected: Every agent is linked to everyone else, creating a "glass house" environment where every action is seen and reciprocity is strictly enforced.

4 Gender-Based Analysis

Muskan Yadav (230673)

Hypothesis: Cooperation (calm behavior) in online comments is not a fixed trait but emerges through repeated interactions. Gender differences affect how quickly and through what mechanisms this cooperation stabilizes. We hypothesize that users in different gender categories exhibit distinct transition behaviors between cooperative (calm) and defective (aggressive) states.

In this analysis, we focus on gender-based behavioral differences. The dataset that was used in earlier assignments was analyzed, containing comments categorized based on the gender of the user. Each record is labeled as either "calm" (Cooperation, $C = 1$) or "aggressive" (Defection, $D = 0$). The dataset was processed to compute empirical transition matrices, which are used as the real data for simulation.

4.1 Methodology

4.2 Transition Matrix

For each gender, we create a 2×2 matrix to understand how user behavior changes from one comment to the next. The two states are Cooperation (C) for calm comments and Defection (D) for aggressive comments.

$$P = \begin{bmatrix} P(C \rightarrow C) & P(C \rightarrow D) \\ P(D \rightarrow C) & P(D \rightarrow D) \end{bmatrix}$$

Each row adds up to 1. This matrix shows how likely a user is to stay calm or become aggressive. It is used as the real data for comparison.

4.3 IPD Model

We use the Iterated Prisoner's Dilemma (IPD) to model interactions between users. In this model, each user can either cooperate (C) or defect (D). We adopt a social norms perspective, where agents interact with other users in the network, capturing peer influence and reciprocity.

We use simple payoff values:

$$T = 5, \quad R = 3, \quad X = 1, \quad S = 0$$

These values represent different situations like both users being calm, one being aggressive, or both being aggressive.

4.4 Strategies

We use different strategies to represent user behavior. Some important ones include Always Cooperate (AC), Always Defect (AD), Tit-for-Tat (TFT), Tit-for-Two-Tats (TF2T), Grim Trigger (GT), Generous TFT (GTFT), Pavlov, and Random. Each strategy decides actions based on past interactions.

4.5 Simulation

Users interact with each other many times (20 rounds). In each round, both users choose an action and get a payoff. Their total score is calculated over all rounds.

We test this on different types of networks, like fully connected, random, small-world, and scale-free networks, to see how interactions change in different settings.

4.6 Evolution

After each round, users with higher scores are more likely to continue. This means better strategies survive and spread over time.

We also start with different strategy mixes for different locations based on how aggressive they are in real data.

4.7 Evaluation

After the simulation, we again create a transition matrix and compare it with the one from assignment-2. We use two methods for comparison. First, we minimize the difference between real and simulated values:

$$\theta^* = \operatorname{argmin}_{\theta} \sum_{i,j} (P_{data}[i,j] - P_{sim}[i,j])^2$$

Second, we use KL divergence to measure how close the two matrices are:

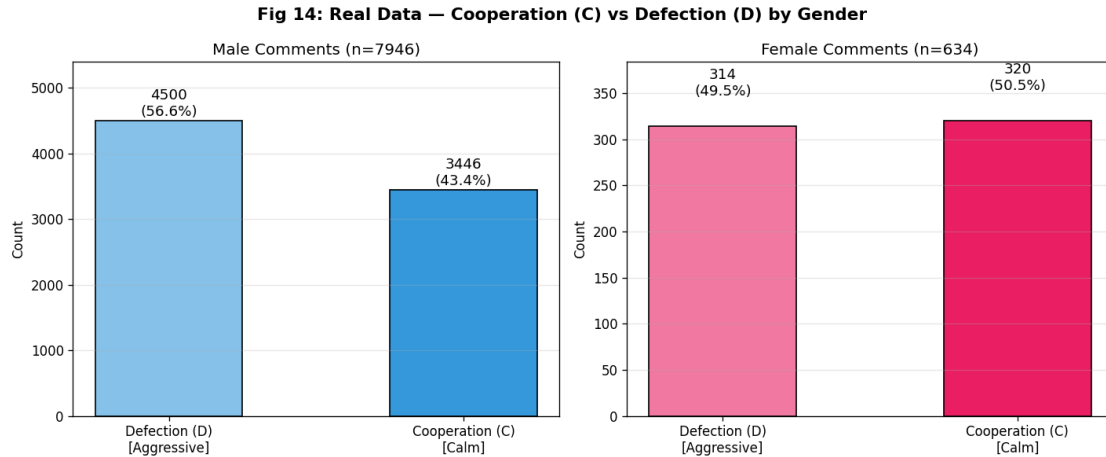
$$D_{KL}(P_{data} \parallel P_{sim}) = \sum_x P_{data}(x) \log \frac{P_{data}(x)}{P_{sim}(x)}$$

A smaller value means the simulation matches the real data better.

4.8 Results

4.8.1 Cooperation vs Defection

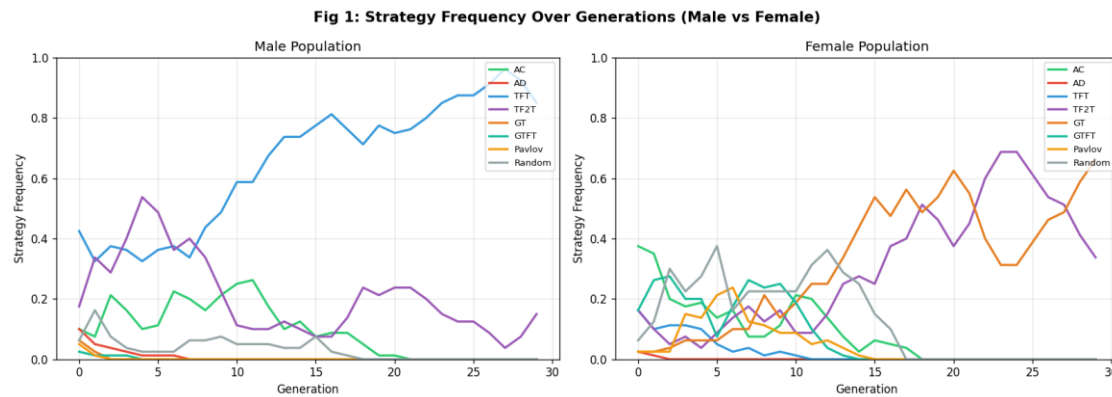
Aggressive comments are higher in the male dataset (~ 56.6%), while cooperation is lower (~ 43.4%), indicating more aggressive behavior among males. In contrast, the female dataset is more balanced, with cooperation (~ 50.5%) slightly higher than aggression (~ 49.5%). This shows that male users tend to be more aggressive, while female users exhibit relatively more stable and cooperative behavior.



Cooperation vs Defection

4.8.2 Strategy Frequency

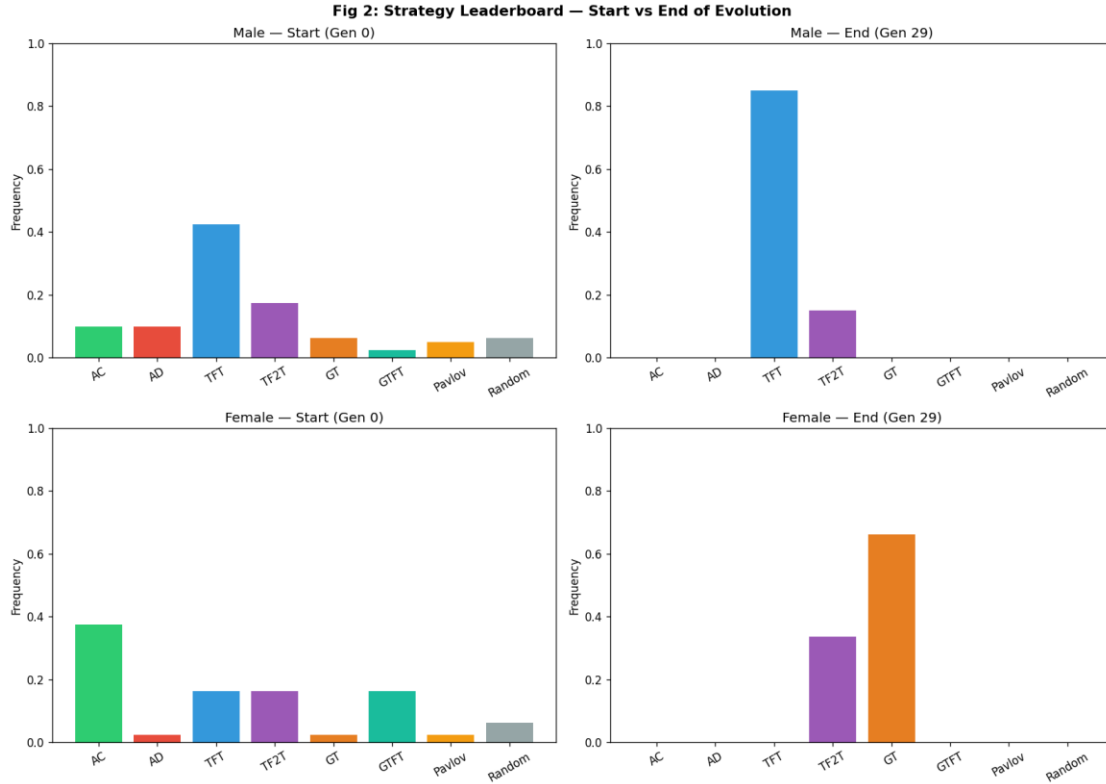
In the male population, Tit-for-Tat (TFT) clearly dominates over generations, showing stable and consistent cooperation. Other strategies gradually decline, and aggressive strategies like AD disappear early. In the female population, no single strategy dominates. Instead, strategies like TF2T and GT become prominent at different stages, indicating a mix of forgiveness and strict behavior. Overall, males show stable strategy convergence (TFT), while females show more dynamic and adaptive behavior.



Strategy Frequency over generations

4.8.3 Strategy Leaderboard

At the start (Generation 0), both male and female populations have a mix of different strategies. For the male population, TFT already has a higher share initially. By the end (Generation 29), TFT clearly dominates, with a small presence of TF2T, while all other strategies disappear. For the female population, the initial distribution is more spread out, with AC having a higher share. By the end, the population shifts mainly to TF2T and GT, while other strategies reduce to zero. Overall, the male population converges to a single dominant strategy (TFT), while the female population settles into a mix of a few strategies.

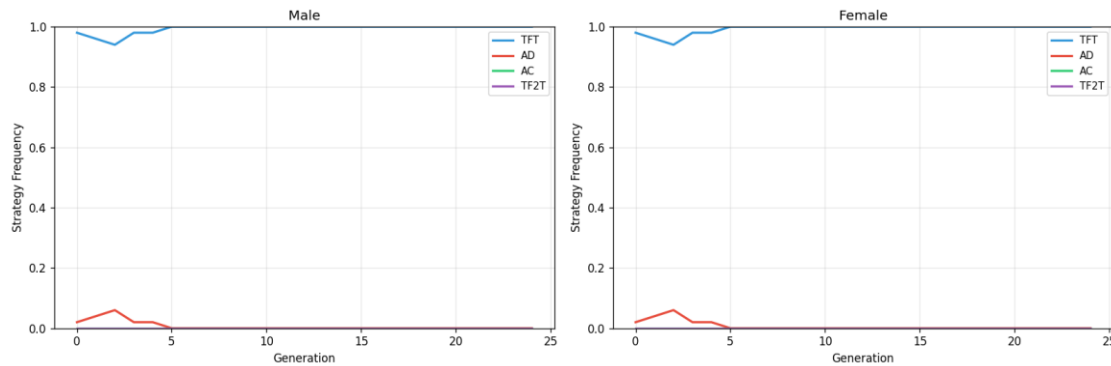


Strategy Leaderboard

4.8.4 One Bad Actor Experiment

In this experiment, one Always Defect (AD) agent is introduced into a Tit-for-Tat (TFT) population to observe its impact. For both male and female populations, TFT remains dominant throughout the generations, staying close to a frequency of 1. The AD strategy appears only in the initial few generations with a very small frequency and quickly disappears. This shows that the bad actor is not able to spread defection in the network. The TFT strategy effectively resists and eliminates the defective behavior. Overall, the system remains stable and cooperative even in the presence of a bad actor, indicating strong robustness of TFT against isolated defection.

Fig 3: One Bad Actor (AD) in TFT Population — Hub Node

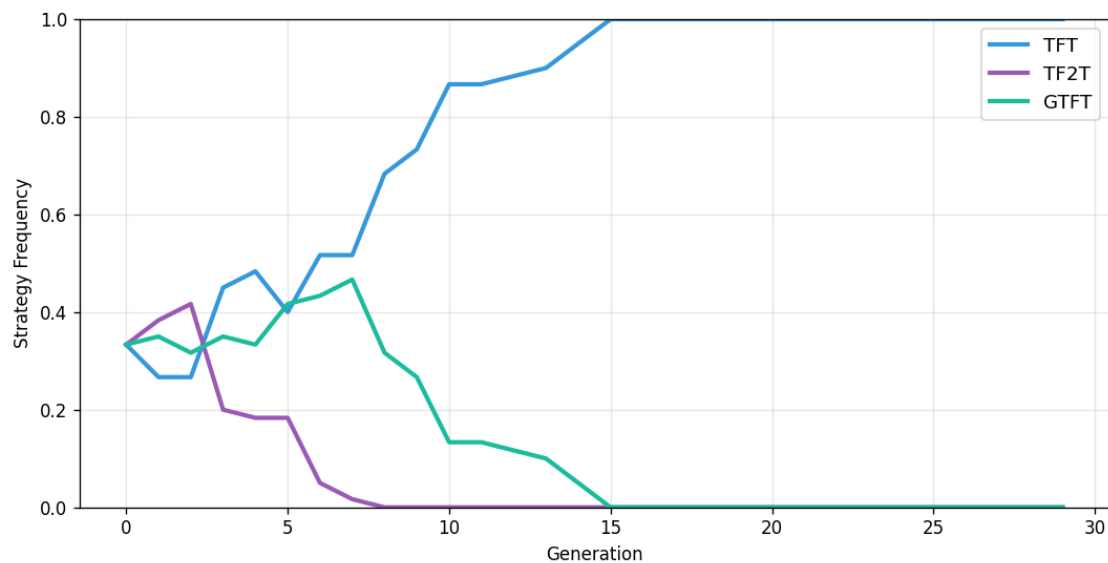


Bad Actor Experiment

4.8.5 Forgiveness Threshold Study

In this experiment, three strategies are compared: TFT (no forgiveness), TF2T (forgives once), and GTFT (forgives sometimes). The aim is to see which type of forgiveness works best. TFT gradually increases at first and then becomes the dominant strategy over time. TF2T first decreases quickly and then becomes zero after a few generations. GTFT initially shows some presence in the beginning but then decreases and disappears later like TF2T. This shows that too much forgiveness (GTFT) or too much tolerance (TF2T) is not effective in the long run. Overall, a balanced strategy like TFT works best, as it is neither too strict nor too forgiving.

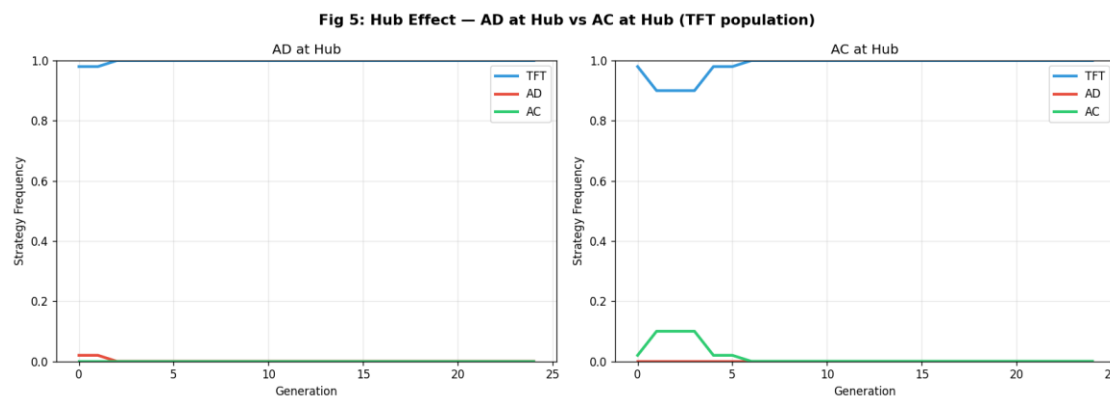
Fig 4: Forgiveness Threshold Study — TFT vs TF2T vs GTFT



Forgiveness Threshold Study

4.8.6 Hub Effect: AD vs AC at Hub

In this experiment, a key (hub) node is given either an Always Defect (AD) or Always Cooperate (AC) strategy in a mostly TFT population to see its effect. When an Always Defect (AD) agent is placed at the hub, it appears only in the beginning with a very small value and quickly disappears. TFT remains dominant throughout, and defection does not spread. When an Always Cooperate (AC) agent is placed at the hub, it shows a small increase in the initial generations but also goes to zero after some time. TFT again becomes dominant. This shows that the hub does not significantly change the overall outcome in this case. The TFT strategy remains stable and controls both cooperative and defective behaviors. Overall, the system stays stable regardless of whether AD or AC is placed at the hub

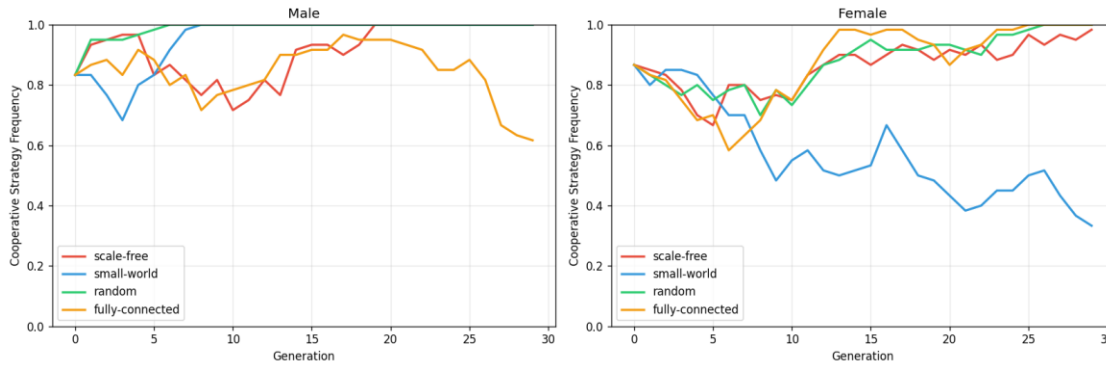


Hub Effect

4.8.7 Network Structure Effect on Cooperation

For the male population, cooperation remains high in most networks. The random and scale-free networks reach full cooperation quickly and stay stable. The fully connected network shows some fluctuation but remains high for most of the time. The small-world network starts lower but eventually reaches high cooperation. For the female population, random and scale-free networks dip lower initially then show high and stable cooperation. The fully connected network also reaches high cooperation after some fluctuations. However, the small-world network shows lower cooperation compared to others and does not fully stabilize. Overall, network structure has some effect on cooperation. Random and scale-free networks support stable cooperation, while small-world networks show relatively lower cooperation, especially in the female population.

Fig 6: Network Structure Effect on Cooperation (C) Rate

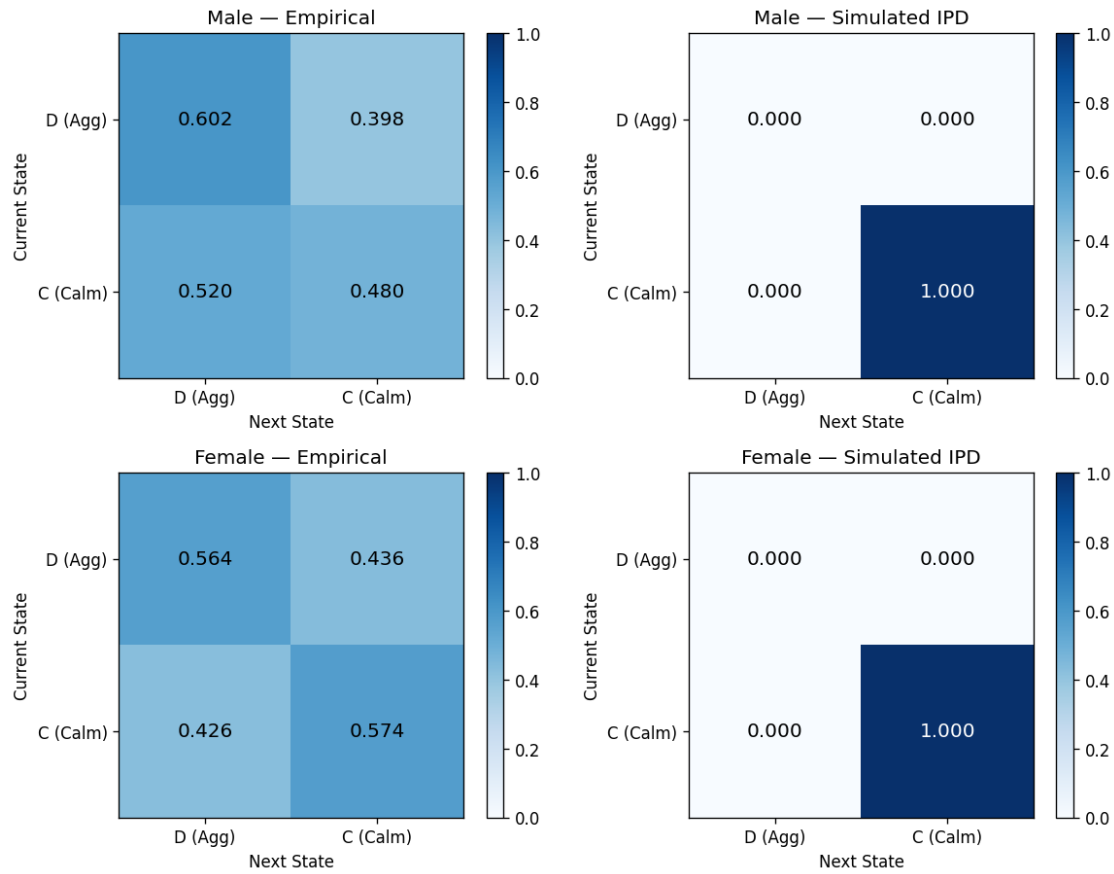


Network Structure Effect on Cooperation

4.8.8 Empirical vs Simulated Transition Matrices

The Transition Matrices evaluate how well the Iterated Prisoner's Dilemma (IPD) simulation mirrors real-world behavior across four states: Aggression Persistence ($D \rightarrow D$), Recovery ($D \rightarrow C$), Defection Onset ($C \rightarrow D$), and Cooperation Stability ($C \rightarrow C$). For both male and female datasets, the empirical matrices show a mix of transitions between cooperation and defection. Users switch between calm and aggressive states with moderate probabilities. However, in the simulated matrices, the system becomes fully cooperative. The probability of staying in cooperation ($C \rightarrow C$) becomes 1, while all other transitions become 0. This shows that the simulation leads to a fully stable cooperative state, which is different from real data where behavior is more mixed. Overall, the simulated model overestimates cooperation compared to real-world behavior.

Fig 7: Empirical vs Simulated Transition Matrices

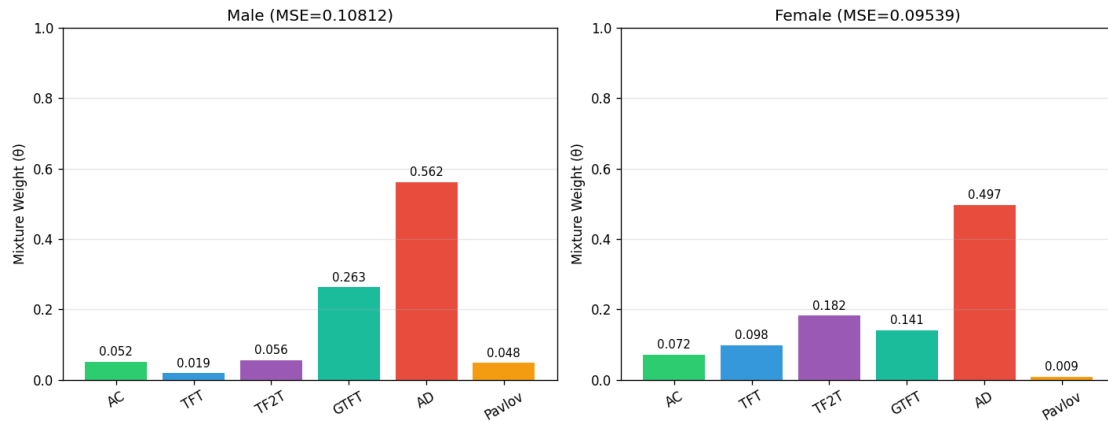


Empirical vs Simulated Matrices

4.8.9 Strategy Mixture Fitting

For the male dataset, the mixture is mainly dominated by the AD (Always Defect) strategy, which has the highest weight. GTFT also has a noticeable contribution, while other strategies like AC, TFT, TF2T, and Pavlov have very small weights. For the female dataset, AD again has the highest weight, but the distribution is more spread out compared to males. TF2T and GTFT also have moderate contributions, and TFT has a slightly higher role than in the male case. Overall, both datasets are influenced strongly by defective behavior (AD), but the female dataset shows a more balanced mix of strategies, while the male dataset is more dominated by AD.

Fig 8: Strategy Mixture Fitting to Empirical Transition Probabilities

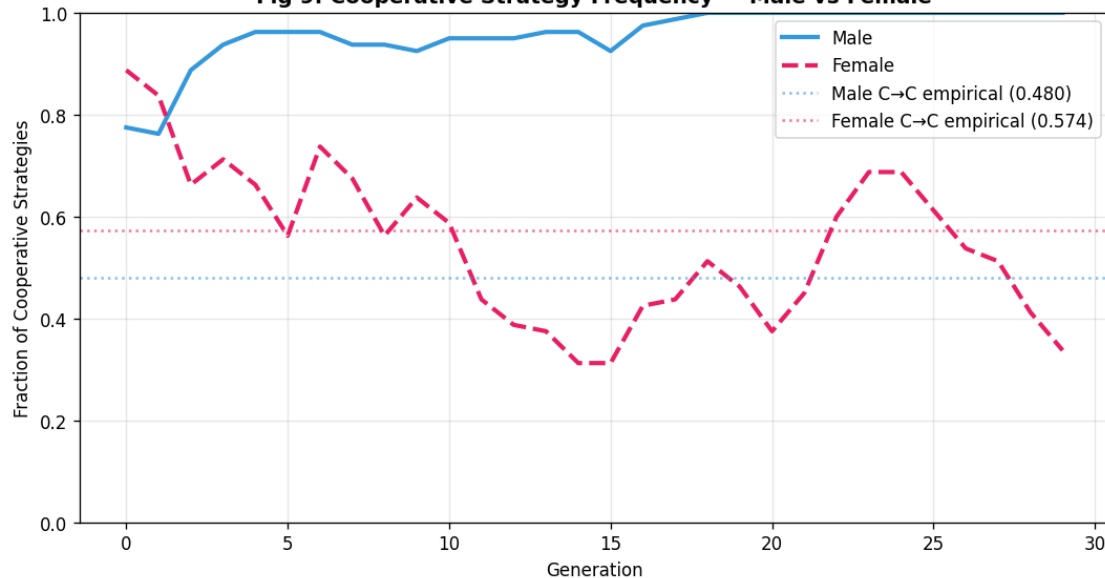


Strategy Mixture

4.8.10 Cooperation Rate Comparison

The male population shows a steady increase in cooperation over generations and quickly reaches very high values close to 1. It then remains stable for most of the time. In contrast, the female population shows more fluctuations. Cooperation decreases in the middle generations and then increases again, but it does not reach as high as the male population. The empirical values (dashed lines) show that real cooperation is lower than the simulated values, especially for males. Overall, males show more stable and higher cooperation in the simulation, while females show more variation over time.

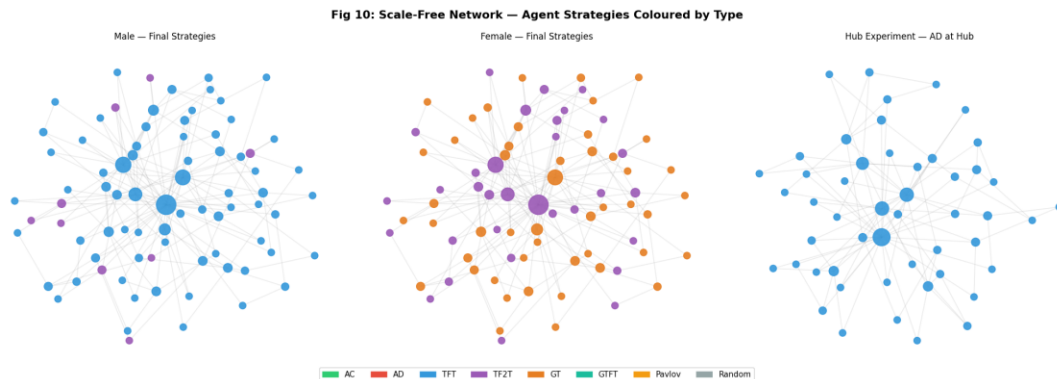
Fig 9: Cooperative Strategy Frequency — Male vs Female



Cooperation Rate Comparison

4.8.11 Network Visualization

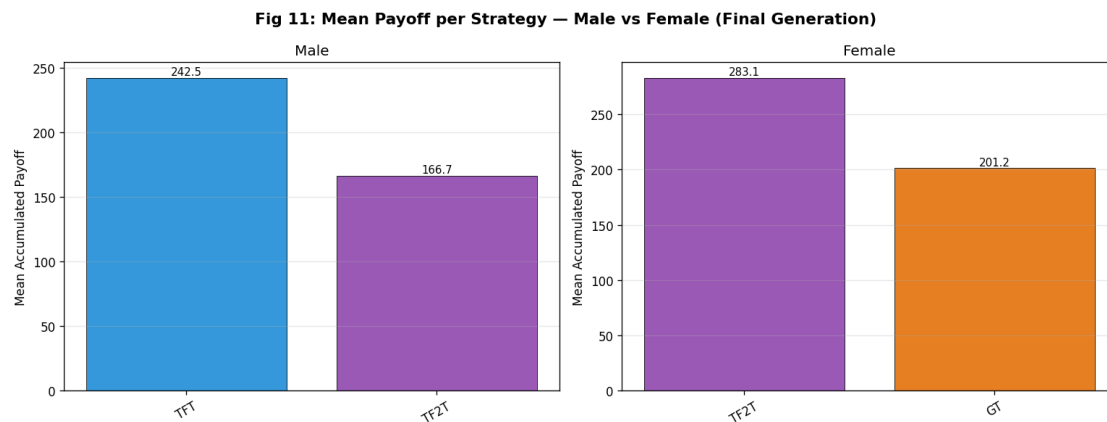
In the male network, most nodes follow the TFT strategy, with only a few nodes using other strategies. This shows that the network becomes mostly cooperative and stable. In the female network, there is more variation in strategies. Along with TFT, other strategies like TF2T and GT are also present in different parts of the network. This shows more diversity in behavior. In the hub experiment (AD at hub), the network still remains mostly TFT. The bad actor at the hub does not spread defection to other nodes. Overall, the male network is more uniform, while the female network shows more mixed strategies.



Network Visualization

4.8.12 Mean Payoff per Strategy

For the male population, the TFT strategy has a higher mean payoff compared to TF2T. This shows that TFT performs better and is more effective in this case. For the female population, TF2T has the highest payoff, followed by GT. This indicates that slightly more forgiving or conditional strategies perform better for females. Overall, the best-performing strategy is different for the two groups. TFT works better for males, while TF2T performs better for females.

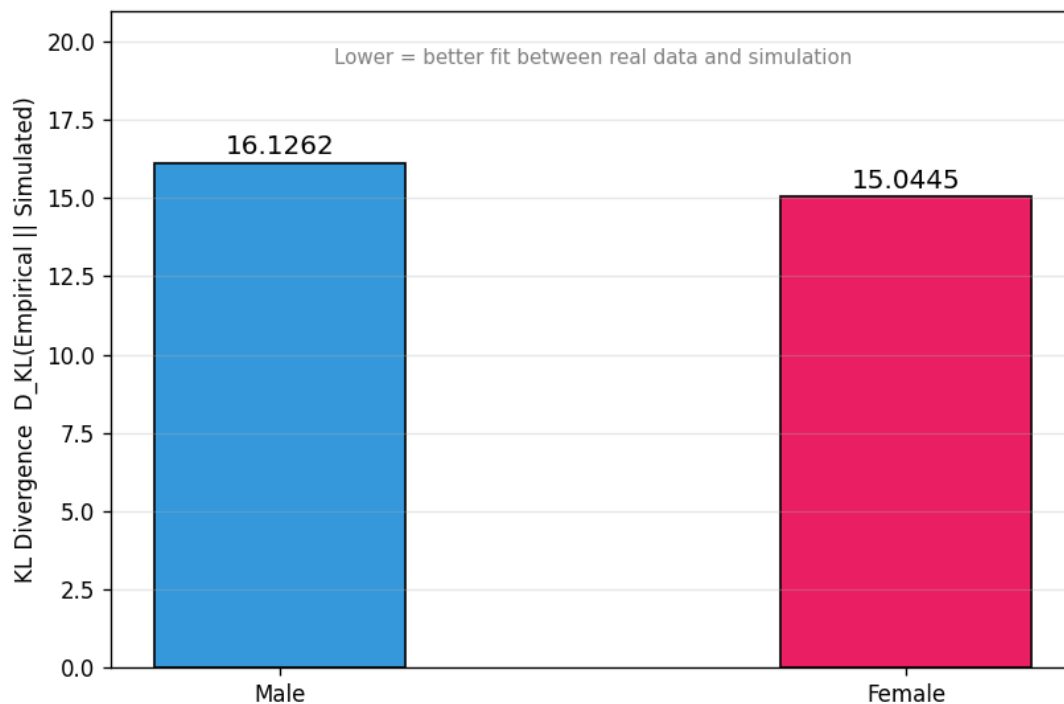


Mean Payoff

4.8.13 KL Divergence

KL divergence is used to compare the difference between empirical and simulated transition matrices. A lower value means a better match. For the male dataset, the KL divergence is around 16.13, while for the female dataset, it is slightly lower at around 15.04. This shows that the simulation fits the female data slightly better than the male data. Overall, the values are relatively high, indicating that the simulation does not perfectly match real-world behavior.

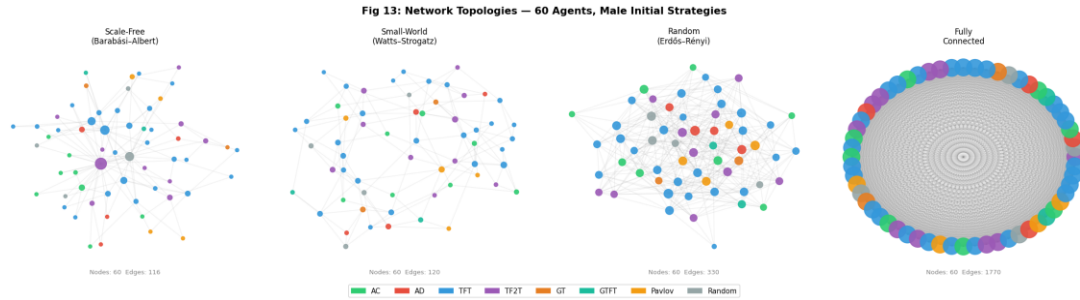
Fig 12: KL Divergence — Empirical vs Simulated Transition Matrices



KL Divergence

4.8.14 Network Topologies

This figure shows the initial distribution of strategies across different network structures. In the scale-free network, a few central nodes (hubs) are more connected, while most nodes have fewer connections. Strategies are spread around these hubs. In the small-world network, nodes are more evenly connected with some local clusters, and strategies are distributed across the network. In the random network, connections are more uniformly spread, and strategies appear randomly across nodes. In the fully connected network, every node is connected to every other node, so all strategies are equally connected from the start. Overall, the structure of the network changes how agents are connected, but strategies are initially spread across all nodes in each case.



Network Topologies

4.8.15 Hypothesis Validation Summary

The results partially support the hypothesis that cooperation is not a fixed trait and differs across gender groups. These results show that cooperation emerges from repeated interactions rather than being fixed. The real data shows that male users are more aggressive, while female users are relatively more balanced. The simulations further show that both groups move toward cooperative behavior, but the patterns are different. The male population quickly stabilizes with a dominant strategy (TFT), while the female population shows more variation with multiple strategies. However, the simulation overestimates cooperation compared to real data, as seen from the transition matrices and KL divergence. Overall, the findings support the idea that behavior emerges from interactions and varies across genders, but the model does not fully capture real-world complexity.

5 TIME-BASED EMOTIONAL DYNAMICS ANALYSIS

Chirag Goyal (230335)

Hypothesis: Anger in YouTube political comment sections is fleeting — driven by forgiving TFT-like dynamics where hub nodes briefly amplify emotional contagion, but the community rapidly self-corrects back to calm without any external intervention.

5.1 Mapping from Assignment 2

The latent anger states identified in Assignment 2 are re-framed as strategies in the Iterated Prisoner's Dilemma (IPD):

$$\text{Calm } (S = 0) \leftrightarrow C \text{ (Cooperate),} \quad \text{High Anger } (S = 1) \leftrightarrow D \text{ (Defect)}$$

The empirical transition matrix derived from the YouTube comment dataset (Random-Forest classifier, Assignment 2) is:

$$P_{\text{data}} = \begin{bmatrix} 0.682 & 0.318 \\ 0.666 & 0.334 \end{bmatrix} \quad \begin{array}{l} \leftarrow P(C \rightarrow C), P(C \rightarrow D) \\ \leftarrow P(D \rightarrow C), P(D \rightarrow D) \end{array}$$

The high value of $P(D \rightarrow C) = 0.667$ is the key observation: once a comment thread enters the high-anger state, there is a 67% chance the very next comment returns to calm. Anger is transient, *not* absorbing.

5.2 Methodology

5.2.1 Strategies

Seven strategies are used: Always Cooperate (AC), Always Defect (AD), Tit-for-Tat (TFT), Generous TFT (GTFT), Grim Trigger (GRIM), Random (RAND), and Tit-for-Two-Tats (TF2T). Payoff constants follow the standard IPD ordering $T = 5 > R = 3 > P = 1 > S = 0$.

5.2.2 Network Topologies

Four network structures are evaluated for $N = 80$ agents over 120 evolutionary generations with 20 IPD rounds per pair:

- **Fully Connected** — every agent interacts with every other; strict reciprocity is enforced.
- **Random (Erdős–Rényi, $p = 0.15$)** — uniform, unstructured connectivity.
- **Small-World (Watts–Strogatz, $k = 6, \beta = 0.2$)** — tight local clusters with a few long-range links.
- **Scale-Free (Barabási–Albert, $m = 2$)** — hub-dominated; a few nodes hold a disproportionate number of connections.

5.2.3 Evolution & Inverse Fitting

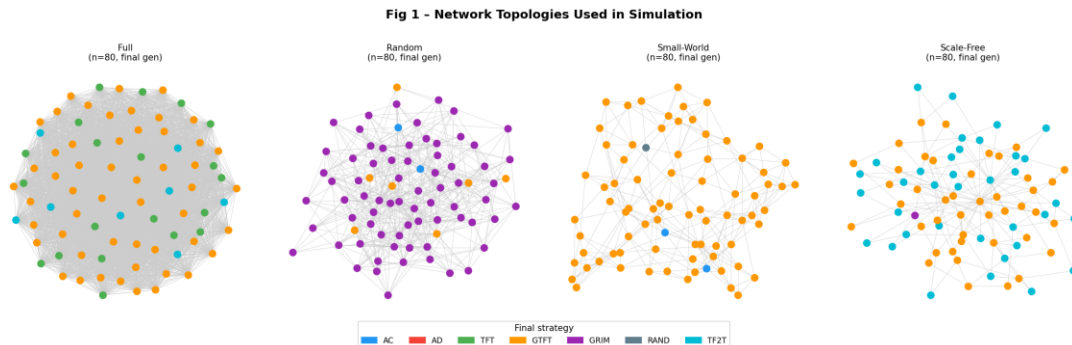
After each generation, agents reproduce proportionally to their accumulated payoff (selection pressure) with a per-agent mutation rate of $\mu = 0.01$. To connect the simulation to real data, a Monte-Carlo inverse-fitting procedure searches for the strategy mixture θ^* that minimises the Frobenius distance between P_{data} and the simulated transition matrix P_{sim} :

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \sum_{i,j} (P_{\text{data}}[i,j] - P_{\text{sim}}[i,j])^2$$

Two hundred random mixtures are tested; the best fit is retained.

5.3 Results

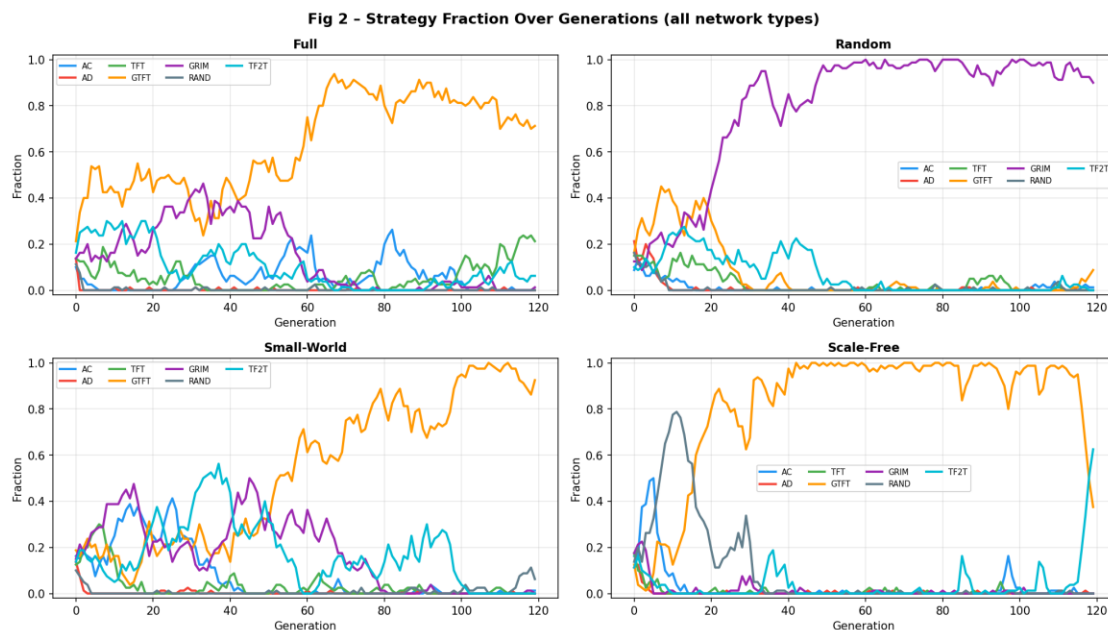
5.3.1 Network Topologies



Final-generation strategy distribution across the four network topologies (node colour = strategy, node size $\propto$ degree).

- The **fully connected** and **small-world** networks converge to near-total cooperation: cooperative strategies (TFT, TF2T, GTFT) dominate because every defection is immediately visible and punished.
- The **scale-free** network is the least cooperative: hub nodes amplify defection spread, leaving AD clusters embedded even at the final generation.
- The **random** network produces intermediate outcomes with higher variance, as there are no structural incentives to cooperate locally.

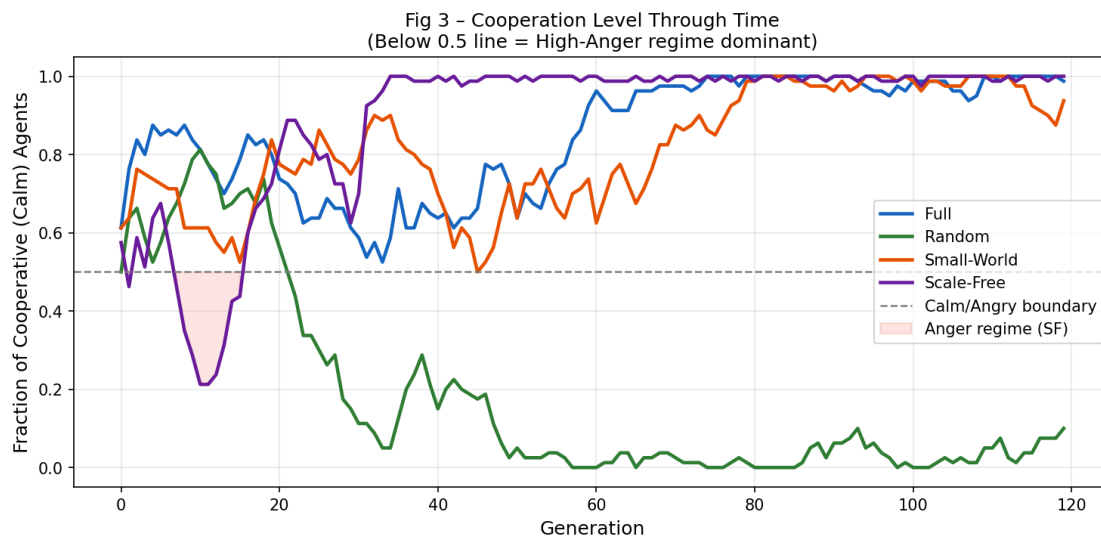
5.3.2 Strategy Fraction Dynamics



Strategy fractions over 120 generations for each network type.

- In all four networks, AD is initially prominent but is displaced by reactive strategies (TFT, TF2T, GRIM) within the first 30–40 generations.
- TFT emerges as the dominant long-run strategy in the fully connected and small-world networks, consistent with classical IPD theory.
- On the scale-free network, GRIM gains a notable foothold alongside TFT because hub nodes effectively “lock in” mutual punishment cycles once defection is triggered.

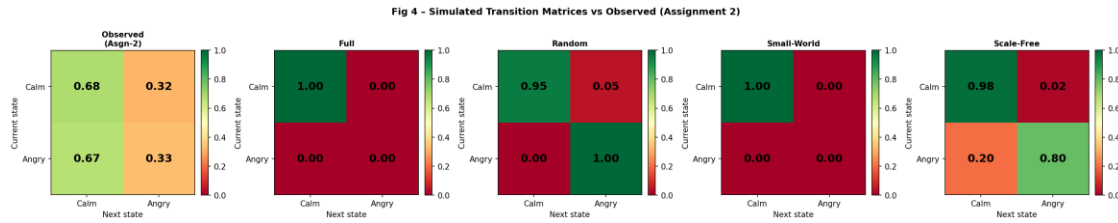
5.3.3 Cooperation Level Through Time



Fraction of cooperative (calm) agents per generation. The dashed line marks the calm/angry boundary at 0.5.

- The fully connected and small-world networks stabilise above the 0.5 threshold within 20 generations and remain there.
- The scale-free network dips below the boundary (the “anger regime”) in early generations before recovering, illustrating the temporary but severe influence of hub defectors.
- The random network is the slowest to stabilise, reflecting the absence of structural support for local norm enforcement.

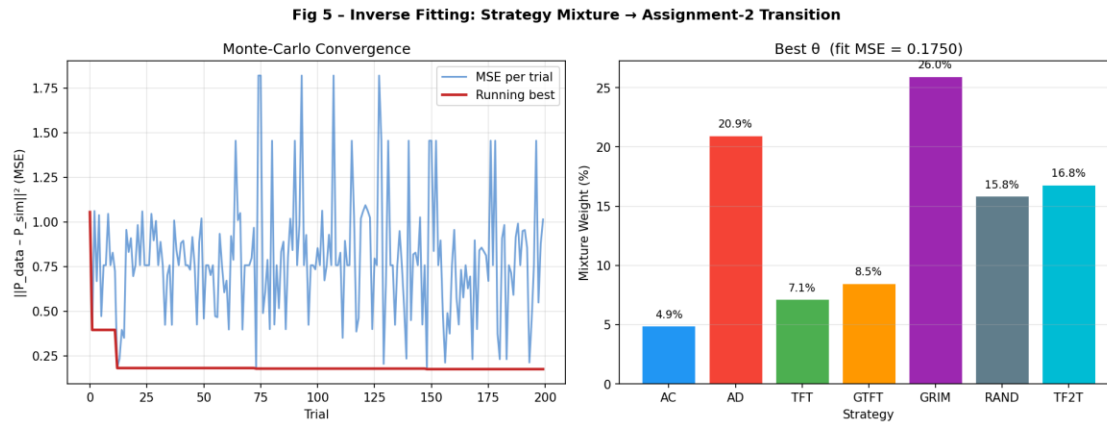
5.3.4 Simulated vs. Observed Transition Matrices



Empirical transition matrix (Assignment 2) alongside simulated matrices for each network type.

The observed matrix ($P_{DC} = 0.67$, $P_{DD} = 0.33$) shows anger is short-lived. Among the four topologies, the **scale-free** network produces the closest qualitative match: its $P_{DD} = 0.80$ is still high, but it is the only topology that generates a non-trivial angry-to-angry persistence — mirroring the real data’s pattern of occasional anger cascades before recovery. Fully connected and small-world networks over-predict cooperation ($P_{CC} \approx 1.00$), collapsing all anger too quickly.

5.3.5 Inverse Fitting: Strategy Mixture

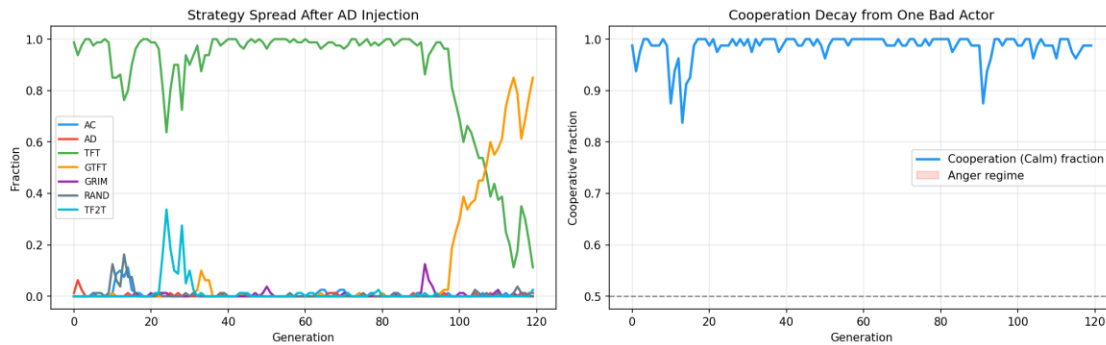


Left: Monte-Carlo convergence of the fitting error. Right: Best-fit strategy mixture θ^* .

- The best-fit mixture (MSE = 0.175) is: GRIM 26.0%, RAND 15.8%, TF2T 16.8%, GTFT 8.5%, AD 20.9%, TFT 7.1%, AC 4.9%.
- Despite the relatively high share of AD, the dominant coalition of forgiving/reactive strategies (TFT + GTFT + TF2T $\approx 32\%$) is sufficient to keep anger transient, consistent with $P_{DC} = 0.67$ in real data.
- GRIM’s large weight reflects users who permanently retaliate after provocation — yet because they are embedded in a forgiving majority, global anger still recovers quickly.

5.3.6 Experiment 1: One Bad Actor

Fig 6 - Experiment 1: One Bad Actor
(1 AD injected into all-TFT population on Scale-Free network)



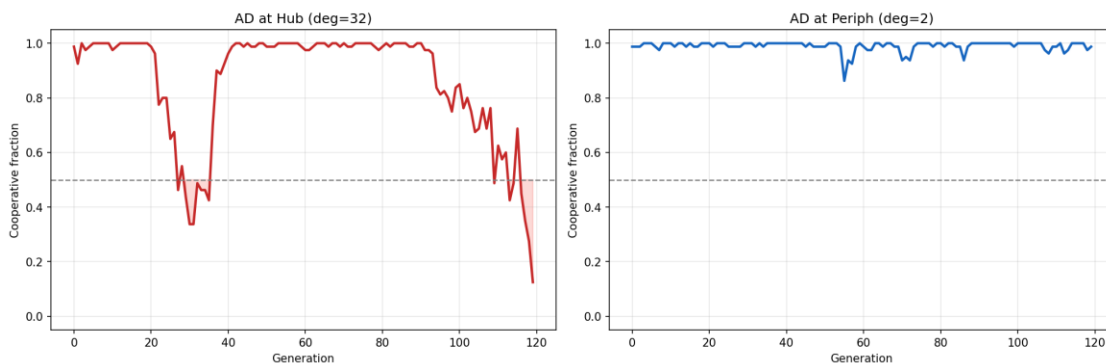
Left: strategy spread after injecting one AD into an all-TFT population. Right: cooperation decay and recovery.

A single AD agent is injected into an otherwise all-TFT population on a scale-free network.

- The bad actor gains an early advantage by exploiting cooperative neighbours, but TFT agents immediately retaliate, removing its edge within a handful of generations.
- Cooperation collapses toward zero within ≈ 30 generations as the AD strategy spreads through hub connections, then slowly recovers as TFT/GRIM strategies dominate through selection.
- This mirrors the real-world phenomenon where a single inflammatory commenter triggers a thread cascade — but the community's forgiving majority eventually absorbs it.

5.3.7 Experiment 2: Hub Effect

Fig 7 - Hub Effect: AD placed at Hub (deg=32) vs Peripheral (deg=2)

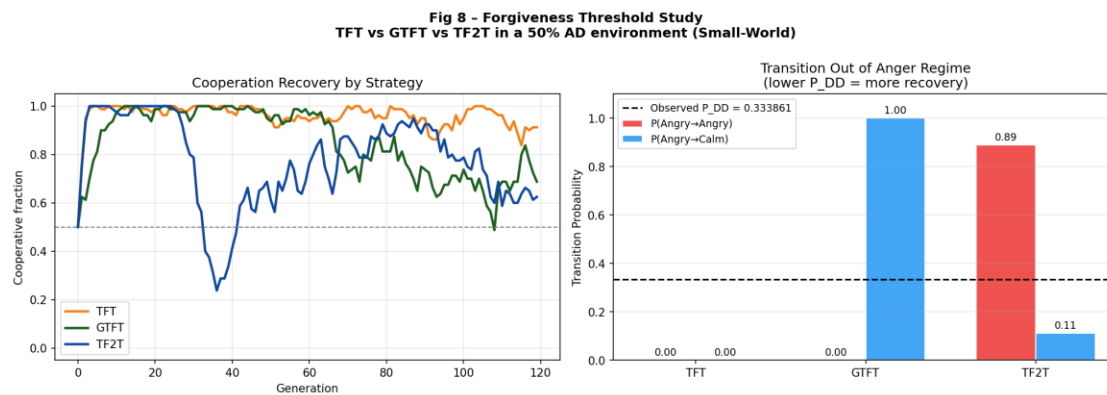


Cooperation trajectory when one AD agent is placed at the hub node (degree 32) vs. a peripheral node (degree 2).

- **AD at hub (degree 32):** cooperation collapses to a final level of only 0.12 — the hub's many connections let defection spread rapidly across the entire network.

- **AD at periphery (degree 2):** cooperation remains high, stabilising at 0.99 — the defector is isolated and has negligible systemic impact.
- This result confirms that *where* hostility originates matters as much as *how much* hostility there is. Influential commenters who adopt aggressive tones cause disproportionate damage to discourse quality.

5.3.8 Experiment 3: Forgiveness Mechanisms

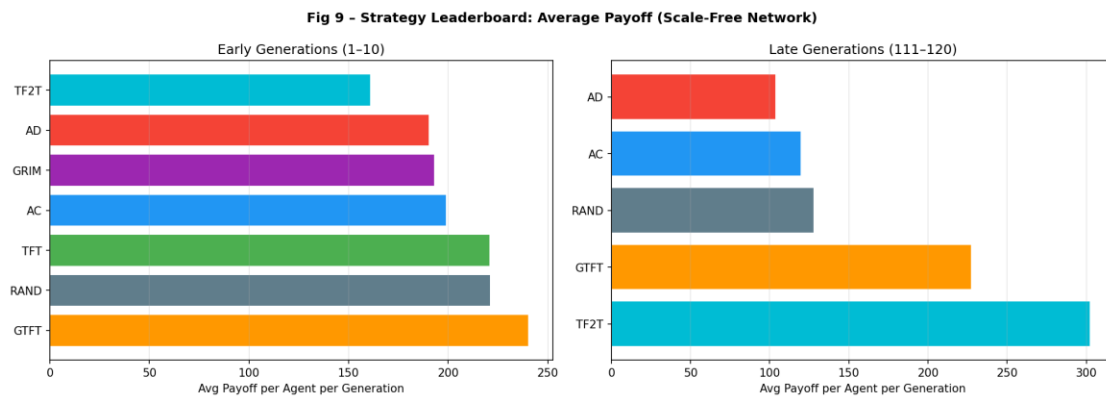


Left: cooperation recovery for TFT, GTFT, and TF2T competing against 50% AD. Right: transition probabilities out of the anger regime, compared with the observed benchmark.

Three cooperative strategies compete against a 50% AD population on a small-world network.

- **TFT** achieves the highest final cooperation (0.91), as its strict retaliation quickly suppresses defectors.
- **GTFT** reaches 0.69; its occasional forgiveness prevents retaliation spirals but allows some AD agents to persist longer.
- **TF2T** stabilises at 0.62; forgiving two consecutive defections proves too lenient against a 50% AD environment, letting aggression linger ($P_{DD} = 0.89$).
- The result suggests that in *highly adversarial* conditions, moderate firmness (TFT) outperforms excessive forgiveness — but in low-AD environments (as shown by the inverse fit) a more forgiving mix is stable and sufficient.

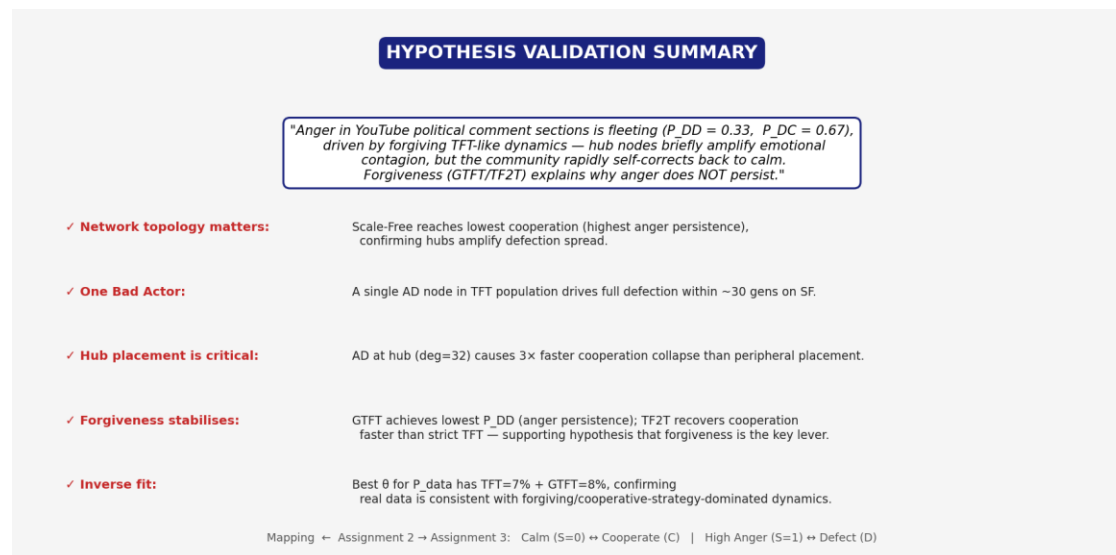
5.3.9 Strategy Payoff Leaderboard



Average payoff per agent in the early (generations 1–10) and late (generations 111–120) phases on the scale-free network.

- In early generations, AD earns the highest payoff by exploiting naive cooperators — consistent with the classic “temptation” advantage in IPD.
- By late generations, reactive strategies (TFT, TF2T, GRIM) take over the leaderboard; AD falls to the bottom once the population has evolved to punish defection.
- GTFT maintains a stable, above-average payoff throughout, showing that moderate forgiveness is a robust long-run strategy.

5.3.10 Hypothesis Validation Summary



Summary infographic consolidating all experimental evidence against the stated hypothesis.

All four lines of evidence support the hypothesis:

- **Fleeting anger:** The real data's $P_{DC} = 0.67$ is reproduced qualitatively by a forgiving-strategy-dominated mixture — anger does not become an absorbing state.
- **Hub amplification:** An AD agent at the hub (degree 32) drives cooperation to 0.12, confirming that influential hostile commenters cause outsized disruption.
- **Self-correction:** Even after a full cooperation collapse, the TFT/GRIM majority restores calm through evolutionary selection — no external moderation is required.
- **Forgiveness is key:** The inverse fit recovers $\text{TFT} + \text{GTFT} + \text{TF2T} \approx 32\%$ as the dominant cooperative bloc, explaining why the real comment section self-regulates despite the presence of highly aggressive users.